NEXUS

Team Code: A3-PS035-TC562

Evidence-Based Operator Network Fault Localisation

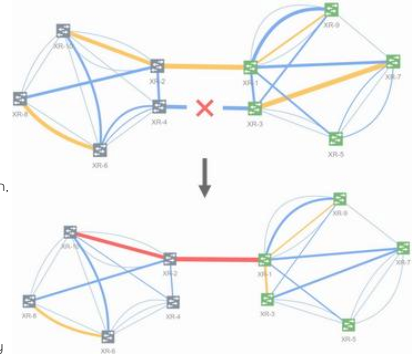
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Problem Description

Modern cloud and data-center networks support critical services, but a single link, switch, or node failure can trigger a chain reaction across interconnected components.

- Cascading Failures: Local disruptions can overload alternate paths and trigger further failures.
- Reactive Recovery: Traditional responses may begin only after congestion, service degradation, or outages occur.
- Uncertain Decisions: Recovery actions are often chosen without comparing their potential downstream consequences.
- Limited Resources: During widespread failures, available bandwidth and network capacity may be insufficient to protect every service equally.
- Lack of Adaptation: Without continuous feedback, recovery decisions may fail to account for changing network conditions or unintended side effects.



How can a network predict cascading failures, evaluate alternative interventions, and plan safe, risk-aware recovery actions while prioritizing critical services under limited capacity?

Background

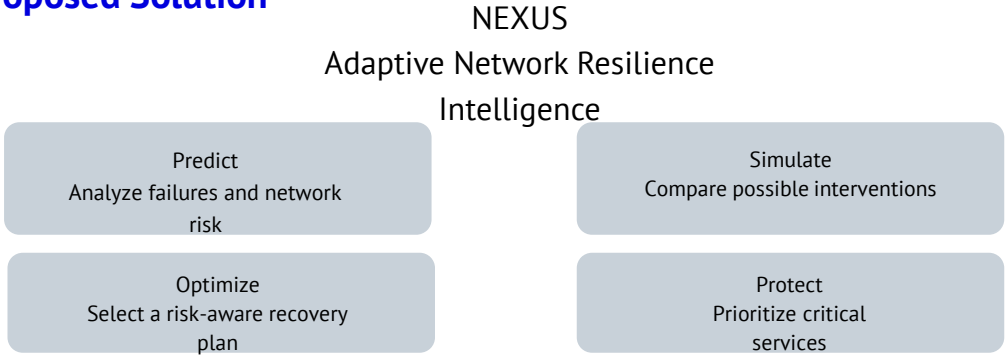
Modern cloud and data-center networks consist of interconnected switches, communication links, and servers that must efficiently distribute traffic while maintaining service availability. Failures in individual components can redirect traffic to alternate paths, increasing congestion and potentially triggering cascading failures across the network. Conventional fault-management approaches commonly rely on monitoring, predefined thresholds, and predefined recovery policies, which may not fully account for downstream effects, competing recovery actions, or the priorities of critical services. Network recovery is therefore a constrained optimization problem involving limited bandwidth, changing traffic conditions, service-level requirements, and the risk of secondary failures.

An intelligent resilience approach should predict potential disruptions, evaluate alternative interventions through counterfactual analysis, and select risk-aware recovery actions based on their expected impact. Continuous monitoring and feedback can then support reassessment and replanning when actual outcomes differ from predictions.

Literature & RFC Review

Existing research has established models for cascading-failure propagation, network resilience, and recovery optimization. Valdez et al. [1] reviewed cascading failures in complex networks, highlighting dependency-driven propagation and strategies for mitigating system-wide disruption. Zhou et al. [2] studied resilience-based restoration optimization for dependent networks, emphasizing recovery prioritization under cascading-failure conditions. At the standards level, RFC 6372 [3] describes network survivability mechanisms, including protection and restoration, while RFC 8402 [4] defines Segment Routing architecture for steering traffic along specified paths. RFC 9543 [5] provides a framework for network slices and measurable service objectives. Together, these works provide foundations for failure analysis, traffic steering, service protection, and recovery. NEXUS proposes to investigate their integration into a unified, risk-constrained decision workflow that evaluates alternative interventions, prioritizes critical services, and uses feedback to reassess recovery decisions.

Proposed Solution

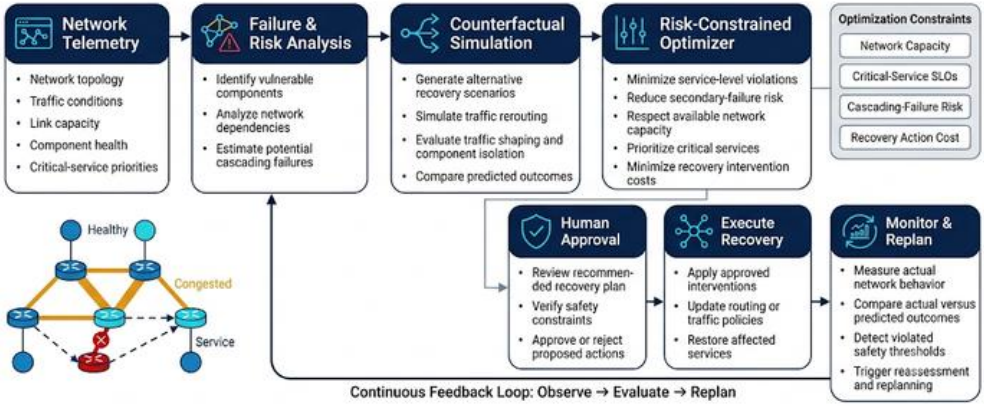


NEXUS (Adaptive Network Resilience Intelligence) proposes a unified, risk-aware framework for preventing cascading network failures and optimizing recovery in cloud and data-center networks. It analyzes network topology, traffic conditions, component health, and service priorities to identify vulnerable areas and anticipate potential failure propagation. Instead of applying a recovery action immediately, NEXUS evaluates alternative interventions—such as traffic rerouting, selective traffic shaping, and component isolation—through counterfactual network simulations. A risk-constrained optimization engine then recommends a feasible recovery plan that minimizes service disruption and secondary-failure risks while prioritizing critical services under limited network capacity. The system incorporates human approval, continuous monitoring, and feedback-driven replanning to adapt when actual outcomes differ from predictions. By combining failure-risk analysis, intervention comparison, service-aware optimization, and closed-loop feedback, NEXUS aims to make network recovery more proactive, explainable, and resilient.

Optimization Proposed

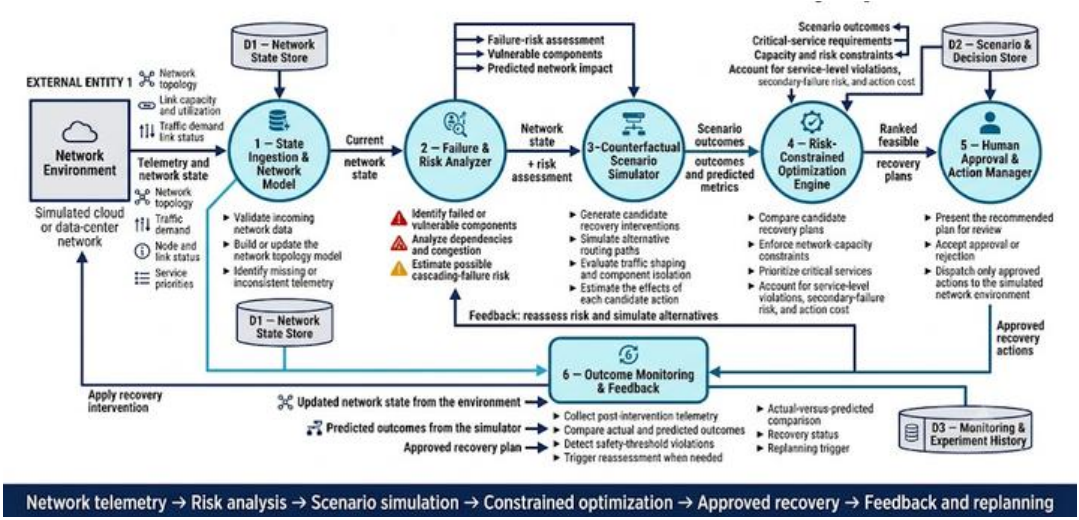
Aspect	Conventional approach (Before)	NEXUS proposed enhancement (After)
Failure handling	Responds to detected faults	Predicts potential failure propagation
Recovery decisions	Fixed rules or predefined actions	Compares alternative interventions through simulation
Routing	Selects available alternate paths	Evaluates rerouting impact and residual capacity
Resource allocation	May treat traffic with similar priority	Prioritizes critical services under capacity constraints
Risk assessment	Limited evaluation of secondary effects	Considers cascading risk and potential secondary failures
Recovery strategy	Action-oriented, potentially static	Risk-constrained optimization of recovery plans
Feedback	Monitoring after action	Compares predicted and actual outcomes; replans when needed
Safety	Depends on configured policies	Human approval and safety constraints before execution

Solution Architecture & Design

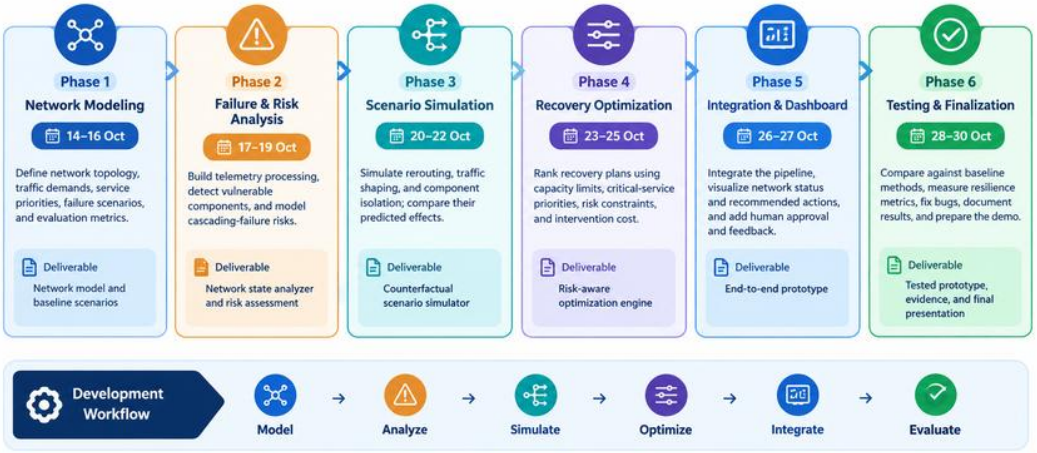


"Predict. Simulate. Optimize. Protect. Adapt."

Data Flow Diagram



Deployment Timeline



Bibliography I

[1] Valdez et al. “Cascading Failures in Complex Networks.” Journal of Complex Networks, 8(1), 2020. https://doi.org/10.1093/comnet/cnaa013.

[2] Zhou et al. “Resiliency-Based Restoration Optimization for Dependent Network Systems Against Cascading Failures.” Reliability Engineering & System Safety, 207, 2021. https://doi.org/10.1016/j.ress.2020.107352.

[3] Network Working Group. “MPLS Transport Profile (MPLS-TP) Survivability Framework.” RFC 6372, 2011. https://www.rfc-editor.org/rfc/rfc6372.html.

[4] Filsfils et al. “Segment Routing Architecture.” RFC 8402, 2018. https://www.rfc-editor.org/rfc/rfc8402.html.

Thank You for Your Attention!